

Grecil Parag Unadkat

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EDUCATION

Vellore Institute of Technology, Chennai Campus

Bachelor of Technology in Computer Science, Specialization in AI and ML

July 2022 – Present

CGPA: 8.67

EXPERIENCE

Machine Learning Intern

Grasim Industries (Aditya Birla Group)

June 2024 – July 2024

Veraval, Gujarat, India

- Developed a chatbot to help employees get information about the company's internal SAP instruction manuals and other SOPs, reducing the daily load on the IT helpdesk by 25%.
- Implemented Pinecone vector database's hybrid search with SPLADE, which almost doubled the accuracy when searching for exact terms, compared to dense vector search.
- Utilised Gemini API for LLM and embeddings, reducing the compute cost by 30% compared to OpenAI.

Competitive Programmer

Google Developer Groups

January 2023 – Present

VIT Chennai, India

- Authored problems for coding contests organized by GDG.
- Delivered a hands-on workshop on binary search to 100+ attendees
- Interviewed more than 30 students as a part of GDG's recruitment drive.

PROJECTS

Corrective RAG | Langchain, Langgraph, Streamlit, FAISS, Gemini API, Docker

June 2024

- Developed an implementation of arxiv.org/abs/2401.15884 using Langchain and Langgraph.
- Outperforms normal RAG in accuracy by 20% on all popular benchmarks and drastically reduces hallucinations.
- Harnesses FAISS in-memory vector database for fast semantic similarity search over indexed data.
- Designed the UI in Streamlit and deployed it using Docker and Streamlit Share.

ZSL based NIDS for Zero Day Attacks | Pandas, Scipy, Tensorflow, Keras, Seaborn

April 2024

- Built a Network Intrusion Detection System inspired by arxiv.org/abs/2109.14868 for the CICIDS2018 Dataset.
- Simulated zero day attack scenario for each attack type by training the model on known attacks and testing on the unknown attack.
- Analysed accuracy, detection rates and F1-scores of all zero day attack scenarios.
- Used Wasserstein distance and Kolmogorov-Smirnov test statistic to find why certain attack types are difficult to detect.

SKILLS

Domains: Deep Learning, Generative AI, Natural Language Processing, Information Retrieval, Software Engineering,

Languages: Python, C/C++, SQL, Neo4j Cypher, Java, HTML/CSS

Frameworks: Flask, FastAPI, Langchain, Tailwind

Developer Tools: Git, VS Code, Docker, Jupyter Notebook, Google Colab, Kaggle, MLflow, Huggingface

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, Scikit-Learn, Tensorflow, Pytorch, Keras, OpenCV, Mediapipe, Transformers, Sentence Transformers

ACHIEVEMENTS

- Earned Reliance Foundation scholarship.
- Selected for ICPC Amritapuri Regionals 2024-25.
- Achieved Expert level on Codeforces(1714), 5 stars on Codechef (2045), Knight on Leetcode (2087).
- Attained global rank 22 in Codechef Starters 175, 392 in Codeforces Round 1025, 411 in Leetcode Biweekly 156.

CERTIFICATIONS

- Foundation Level in Programming and Data Science from IIT Madras.
- Neo4j Certified Professional from Neo4j GraphAcademy.
- Networking Basics by Cisco.